Bibliographic synthesis of the extended maritime power formula

Attribution chain by component · author · publication · contribution to the model · verified pages

$$MP_{norm} = [(MI \times NP) \times SW \times MS^{\gamma} \times CP^{\alpha}]^{1/(3+\gamma+\alpha)} \quad \cdot \quad all\ variables \in (0, 1]$$

Ecs. (1)–(7) · Normalization: geometric mean — PNUD (2010); OCDE (2008) · Colombia: $MP_{norm} \approx 0.53$ — anchored to the scale of Vargas Suárez et al. (2021, p.295)

PM CLASSICAL CORE · MI×NP

`Base preserved in (MI×NP)`

Solís Oyarzún	1997	*Strategy manual. ACANAV* · PM=IM×PN; T.I p.246, T.II p.10
Till, G.	2007	*Sea Power. IPN* · tight relation MI ↔ PN (Uribe et al., 2016, p.47)
Uribe Cáceres et al.	2016	*Estrategia marítima. ESDEG* · Fig.5 p.47; CM p.64; VE p.68; PN pp.77-78
Uribe Cáceres (Ed.)	2021	*Estrategia marítima (2.ª ed.)* · Mahan genealogy (pp.27-28, 115) y Corbett
González Núñez	2017	*Ensayos EM, 5, 31-42* · ARC institutional adoption (p.34)
Schnaidt Mecklenb.	2005	*REVISMAR, (3), 237-250* · "MP composed of NP and MI" (p.242)
ARC	2022	*Plan de Desarrollo Naval 2042* · strategic objective MI+NP (p.58)

IM MARITIME INTERESTS

$MI = \Sigma w_i \cdot MI_i \ ; \ MI \in (0,1]$

Till, G.	2007	*Sea Power. IPN* · State motivations regarding the sea; prior identification
Uribe Cáceres et al.	2016	*Estrategia marítima. ESDEG* · MI as the purpose of MP (p.47)
CCO	2021	*Colombian maritime interests.* · n=18 MI of Colombia; weights w_i
OCDE	2008	*Handbook Composite Indicators* · weighted aggregation methodology ec.(2)
Vargas Suárez et al.	2021	*RCGJMC, 19(34), 267-306* · Colombia MP rank 33/99; index 0.44 (p.295)

PN NAVAL POWER

$NP = F \times P \times MDA^{\mu} \ ; \ NP \in (0,1]$

Mahan, A. T.	1890	*Influence of Sea Power* · classical basis of naval power
Corbett, J. S.	1911	*Principles of Maritime Strategy* · command of the sea and strategy
Castex, R.	1931	*Théories stratégiques* · external factors of strategy
Till, G.	2007	*Sea Power. IPN* · synthesis of force and position; technology and doctrine
Uribe Cáceres et al.	2016	*Estrategia marítima. ESDEG* · Colombian NP: platforms, HR (pp.77-78)
Bebber, R. J.	2022	*USNI Proceedings, 148(12)* · MDA at structural level F×P; CP enables NP
Cabuya-Padilla	2024	*PhD dissertation. ENAP-ENP* · tactical CD in F; tactical vs. strategic distinction
Martínez et al.	2024	*Int. J. Inf. Security, 23, 1429* · MDA systems recurrently compromised (AIS·GNSS·ECDIS)

VE STRATEGIC WILL

$SW = (PW^{w_1} \cdot IW^{w_2} \cdot SocW^{w_3})^{1/\Sigma w}$

Uribe Cáceres et al.	2016	*Estrategia marítima. ESDEG* · SW dynamizer of NP (p.68); MC def. (p.64)
Alba-Rocha et al.	2022	*RISTI, (49), 211-221* · conceptual model of the Maritime Power System at strategic level
Albarán Gómez	2019	*Justicia, 24(35), 13-28* · naval power for development; realism to cooperation
Mora Peña & Uribe C.	2020	*Ensayos EM, 4(11), 14-23* · pillars of Colombia as a bioceanic power
Till, G.	2007	*Sea Power. IPN* · political-social consensus; MP sustainability
PNUD	2010	*Informe Desarrollo Humano* · geometric mean HDI → avoids compensation PW/IW/SocW
OCDE	2008	*Handbook Composite Indicators* · non-substitutable dimensions → ec. (4)
Realpe Díaz	2024	*Novum Jus, 18(2), 279-304* · CDip as expression of PW; dual role (pp.297-299)

SM MARITIME SECURITY

$SM = (CI^{a_1} \cdot TE^{a_2} \cdot BC^{a_3})^{1/\Sigma a}$

Bueger & Edmunds	2024	*Understanding Maritime Security. OUP* · CI, TE, BC; MDA transversal
OMI	2021	*Guide Maritime Security & ISPS* · ISPS, SUA; regional frameworks (Yaundé, ReCAAP)
OMI	2017	*Resolución MSC.428(98)* · cyber risk management (2017); enforceable since 2021
Vargas Suárez et al.	2021	*RCGJMC, 19(34), 267-306* · Colombia MS rank 75/99 (0.59); Group B (p.295)
Fahim et al.	2021	*IEEE Eng. Mgmt. Rev., 49(4)* · ports and physical internet; infrastructure resilience

CP CYBER POWER — CENTRAL CONTRIBUTION OF THE ARTICLE

$CP = \min\{1, (\beta_1 \cdot CS_m + \beta_2 \cdot CD_m + \beta_3 \cdot CR_m) \cdot (1 + \delta \cdot CDb)\}$ · $\alpha(IMCE) < 1$ → concave: initial returns ↑

Nye, J. S.	2010	*Cyber Power. Belfer Center* · canonical def. of CP; resources vs. outcomes
Caro Bejarano	2013	*Pre-bie3, (3), 7. IEEE España* · hard vs. soft CP distinction applicable to maritime CP
Bebber, R. J.	2022	*USNI Proceedings, 148(12)* · CP = key element of MP; CP dependencies ↔ PM
Espinel Bermúdez	2022	*Ensayos EM, 6(16), 79-95* · MP cyber dependence Col.; Maersk/USCG (p.89)
Lagouvardou	2018	*MSc thesis. DTU* · IT/OT separation as the main domain gap
Cabuya-Padilla	2024	*PhD dissertation. ENAP-ENP* · MARCIM: CS_m, CD_m, CR_m; α(IMCE)
Cabuya-P. & Castaneda-M.	2024	*DYNA, 91(231)* · LATAM state of the art; +900% cyberattacks (p.171)
Cabuya-Padilla et al.	2025	*Int. J. Inf. Security, 24, art.122* · SERDUX-MARCIM; Maersk 300M USD/10 days
Martínez et al.	2024	*Int. J. Inf. Security, 23, 1429* · POSEIDON; AIS/ECDIS/GMDSS; IT/OT convergence
Realpe Díaz	2024	*Novum Jus, 18(2), 279-304* · CDip amplifies CP; coef. δ (pp.297-299)
Realpe & Ayala	2026	*Novum Jus, 20(2), 79-105* · CP = technical/normative/defensive/diplomatic capabilities → digital sovereignty
ONU	2021	*A/76/135 — Inf. GEG ICTs* · 11 voluntary norms; reference CDip+PW
UIT	2024	*Global Cybersecurity Index (5.ª ed.)* · Colombia level 3 "consolidating"; α≈0.4–0.7
Cano, J. J.	2024	*Revista Sistemas, (170)* · data-driven cyber defense; conceptual model
Suarez Medina	2024	*Rev. Ciberespacio, 3(5)* · sophisticated cyber threats (interview with Cano)
Realpe & Cano	2020	*CIBSI 2020. U. Rosario* · threats to national security and defense
DNP · CONPES	2020	*CONPES 3995* · National Policy on Digital Trust and Security
IT/OT cases	2019–24	*Maersk · USCG* · incidents: Wesley, Abbatemarco, USCG, Akpan et al.

n_{eff} NORMALIZATION · EQS. (1B-1C)

`MP_norm = [PM]^(1/(3+γ+α)) · MP_norm∈(0,1]`

PNUD	2010	*Informe Desarrollo Humano* · methodological precedent: HDI with geometric mean
OCDE	2008	*Handbook Composite Indicators* · normalización non-substitutable dimensions; ec.(1b-1c)

⊕ TRANSVERSAL METHODOLOGICAL DECISIONS

MDA	Not an independent variable → distributed across NP(F), MS, and CP · Bebber (2022); Bueger & Edmunds (2024)
MC ≠ MDA	MC: cultural-identity (Uribe et al., 2016, p.64) · MDA: técnico-operacional · transversal a VP·IW·SocW
CDip dual role	Projection coef. in CP + expression of PW in SW · Realpe Díaz (2024, pp.297-299)